Affine Invariant Samplers and Flows

analysis and new algorithms

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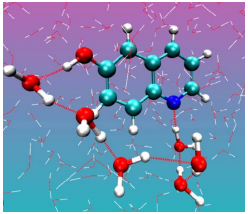
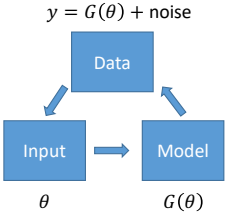
Dynamical sampling and MCMC

Sampling from probability distributions $\pi \propto \exp(-V)$ has been a classical and fundamental challenge in scientific computing and statistics

Applications in

- Statistical physics
- Bayes inverse problems
- Filtering
- Uncertainty quantification
- ...





Markov chain Monte Carlo (MCMC) has been the workhorse for sampling

Typically, at a discrete m-step: $\mathbf{x}(m+1) = R(\mathbf{x}(m), \pi)$

where R is some random mapping. As $m \to \infty, \mathbf{x}(m)$ approximately drawn from π

Challenges of anisotropy and affine invariance

Probability distributions can often be highly anisotropic or ill-conditioned, such as due to hierarchical Bayes or multiple scales in physical fields

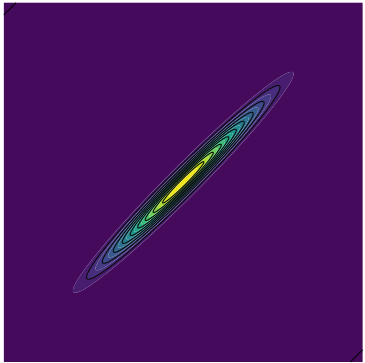


Figure: $\exp\left(-\frac{(x_1-x_2)^2}{2\epsilon} - \frac{(x_1+x_2)^2}{2}\right)$

- Ordinary MCMC may suffer from anisotropy; need preconditioning and adaptation
- **Affine invariance**: for any invertible affine transformation $\mathbf{y} = \phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$

$$\mathbf{x}(m+1) = R(\mathbf{x}(m), \pi)$$

$$\Leftrightarrow \mathbf{y}(m+1) = R(\mathbf{y}(m), \phi\#\pi)$$

[Goodman, Weare 2010]

- Concept motivated by affine invariance of optimization algorithms: Nelder-Mead simplex [Nelder, Mead 1965] and Newton

Affine invariant ensemble samplers by Goodman-Weare

Ensemble samplers with **affine invariance**
J Goodman, J Weare - … in applied mathematics and computational science, 2010 - msp.org
… of a practical **sampler** that has this **affine invariance** property for any general class of densities. In this paper we propose a family of **affine invariant ensemble samplers**. An **ensemble**, X, …
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emcee: the MCMC hammer
D Foreman-Mackey, DW Hogg, D Lang… - Publications of the …, 2013 - iopscience.iop.org
We introduce a stable, well tested Python implementation of the affine-invariant ensemble sampler for Markov chain Monte Carlo (MCMC) proposed by Goodman & Weare (2010). The …
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emcee package routinely used in applications, especially in astrophysics

e.g., Dark energy [Riess et al. 2022], Extrasolar planets [Fulton et al. 2017]

Affine invariant ensemble sampler algorithm

Algorithm is simple: given a collection of walkers $(\mathbf{x}_1, ..., \mathbf{x}_N)$ targeting at π^N

- Stretch move randomly selects two $\mathbf{x}_i$ and $\mathbf{x}_j$

$$\tilde{\mathbf{x}}_i = \mathbf{x}_j + Z(\mathbf{x}_i - \mathbf{x}_j)$$

  where density of Z satisfies $g(z) \propto \frac{1}{\sqrt{z}}$ for $z \in [a^{-1}, a]$; recommended $a = 2$. Parallel version available

- Accept this proposal $\mathbf{x}_i \to \tilde{\mathbf{x}}_i$ with probability

$$\min\{1, Z^{d-1}\frac{\pi(\tilde{\mathbf{x}}_i)}{\pi(\mathbf{x}_i)}\}$$

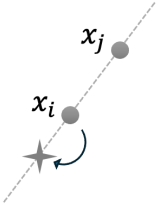

Illustration of stretch move

- Affine invariance since all depend on relative locations

Affine-invariant ensemble sampler is reported to suffer from high dims (e.g. $d \geq 50$) [Huijser, Goodman, Brewer 2015], and theory largely underdeveloped

Outline of This Talk

1. Continuous scaling limit as *affine invariant gradient flows*
2. *Unbiased* affine invariant *gradient* samplers with improved dimensional scaling
3. *Approximate* variational inference with affine invariant gradient flows

Continuous scaling limit at stationarity

Theorem [Chen 2026+] (informal): For $\delta > 0$, track the stretch move single-particle chain $(\mathbf{x}_n^\delta)_{n\geq 0}$ at stationarity, with $a_\delta = 1 + \ell\sqrt{\delta}$. Rescale time by $t = n\delta$ and define the continuous-time interpolation $\mathbf{X}_t^\delta := \mathbf{x}_{\lfloor t/\delta \rfloor}^\delta$. Then, for every finite $T > 0$,

$$\mathbf{X}^\delta \implies \mathbf{X} \quad \text{in } D([0,T],\mathbb{R}^d) \quad \text{as } \delta \to 0$$

where $\mathbf{X}$ satisfies

$$\mathrm{d}\mathbf{X}_t = \tfrac{1}{2} C(\mathbf{X}_t)\nabla \log \pi(\mathbf{X}_t)\,\mathrm{d}t + \tfrac{1}{2}(\nabla \cdot C)(\mathbf{X}_t)\,\mathrm{d}t + C(\mathbf{X}_t)^{1/2}\,\mathrm{d}\mathbf{W}_t$$

Here $C(\mathbf{x}) = \sigma^2\,\mathbb{E}_{\mathbf{x}'\sim\pi}\left[(\mathbf{x}-\mathbf{x}')(\mathbf{x}-\mathbf{x}')^\top\right]$ and $\sigma^2 = \frac{\ell^2}{3}$

Proof using similar techniques in diffusion limits [Roberts, Gelman, Gilks 1997], [Roberts, Rosenthal 1997], etc.

The limiting dynamics is a *gradient flow* under $C(\mathbf{X})$-weighted Wasserstein metric

Gradient flows over probability space

Formally: let $\mathcal{P}$ denote the space of probability densities

- given an energy functional $\mathcal{E} : \mathcal{P} \to \mathbb{R}$
- and a metric $g_\rho : T_\rho\mathcal{P} \times T_\rho\mathcal{P} \to \mathbb{R}$, $g_\rho(\sigma_1, \sigma_2) = \langle M(\rho)\sigma_1, \sigma_2\rangle_{L^2}$. Here tangent space $T_\rho\mathcal{P}$ consists of measures integrated to 0

$$\Longrightarrow \text{gradient flow:} \quad \frac{\partial \rho_t}{\partial t} = -\nabla_g \mathcal{E}(\rho_t) = -M(\rho_t)^{-1} \frac{\delta \mathcal{E}}{\delta \rho}|_{\rho=\rho_t}$$

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Classical example: with $\mathcal{E}(\rho; \pi) = \mathrm{KL}[\rho \| \pi] = \int \rho \log \frac{\rho}{\pi}$ for which $\frac{\delta \mathcal{E}}{\delta \rho} = \log \frac{\rho}{\pi} + 1$

$$\begin{aligned}
\text{Metric:} \quad & M(\rho)^{-1}\psi = -\nabla \cdot (\rho \nabla \psi) \\
\text{Flow:} \quad & \frac{\partial \rho_t}{\partial t} = -\nabla \cdot (\rho_t \nabla \log \pi) + \nabla \cdot (\nabla \rho_t) \\
\text{SDE:} \quad & \mathrm{d}X_t = \nabla \log \pi(X_t)\mathrm{d}t + \sqrt{2}\mathrm{d}W_t
\end{aligned}$$

[Jordan, Kinderlehrer, Otto 1998], [Otto 2001], [Ambrosio, Gigli, Savaré, 2005], etc.

Affine invariant gradient flows for sampling π

Definition: The following gradient flow is called *affine invariant*

$$\frac{\partial \rho_t}{\partial t} = -M(\rho_t)^{-1} \frac{\delta \mathcal{E}}{\delta \rho}(\rho; \pi)\Big|_{\rho=\rho_t}$$

if under the invertible affine transformation $\tilde{x} = \varphi(x) = Ax + b$, it holds that

$$\frac{\partial \tilde{\rho}_t}{\partial t} = -M(\tilde{\rho}_t)^{-1} \frac{\delta \mathcal{E}}{\delta \rho}(\tilde{\rho}; \tilde{\pi})\Big|_{\tilde{\rho}=\tilde{\rho}_t}$$

where $\tilde{\rho}_t = \varphi \# \rho_t$ and $\tilde{\pi} = \varphi \# \pi$

[Chen, Huang, Huang, Reich, Stuart 2023]

- Convergence of ρ_t towards π mirrors convergence of $\tilde{\rho}_t$ towards $\tilde{\pi}$
- Thus automatically adapt to the optimal affine preconditioned coordinates

Examples of affine invariant gradient flows

There are many affine invariant gradient flows

- Covariance preconditioned Wasserstein gradient flow, using

$$M(\rho)^{-1}\psi = -\nabla \cdot (\rho \mathrm{Cov}(\rho)\nabla\psi)$$

  [Reich, Cotter 2015], [Garbuno-Inigo, Hoffmann, Li, Stuart 2020], [Garbuno-Inigo, Nüsken, Reich 2020], ...

- $C(\rho, \mathbf{x})$-weighted Wasserstein gradient flow, using

$$M(\rho)^{-1}\psi(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot (\rho(\mathbf{x})C(\rho, \mathbf{x})\nabla_{\mathbf{x}}\psi(\mathbf{x}))$$

- Fisher-Rao gradient flow: using

$$M(\rho)^{-1}\psi = \rho(\psi - \mathbb{E}_\rho[\psi])$$

  it leads to the unique *diffemorphism invariant* gradient flow
  [Rao 1945], [Amari 2016], [Ay, Jost, Lê, Schwachhöfer, 2017], ...

- Affine invariant Stein flow, ... (see paper)

Unbiased samplers with affine invariant Wasserstein gradient flows as proposal moves

Covariance preconditioned Langevin as proposals in Metropolis-Hastings:

1. given a collection of walkers $\mathbf{x}_1, \ldots, \mathbf{x}_N$ targeting π^N
2. divide them to two groups (for parallelism)

$$S^{(0)} = \{\mathbf{x}_1, ..., \mathbf{x}_{N/2}\},\ S^{(1)} = \{\mathbf{x}_{N/2+1}, ..., \mathbf{x}_N\}$$

3. fix walkers in $S^{(1)}$. For $1 \leq i \leq N/2$, propose (in parallel)

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i + h B_{S^{(1)}} B_{S^{(1)}}^T \nabla \log \pi(\mathbf{x}_i) + \sqrt{2h} B_{S^{(1)}} \mathbf{z}_i$$

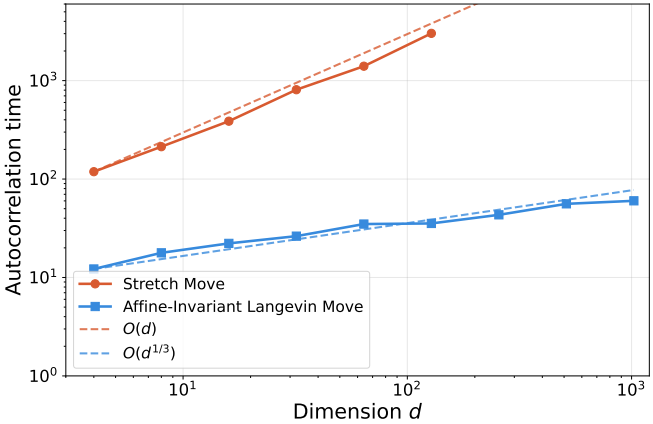
   where $B_{S^{(1)}} = \frac{1}{\sqrt{N/2}}[\mathbf{x}_{N/2+1} - \mathbf{m}_{S^{(1)}}, \ldots, \mathbf{x}_N - \mathbf{m}_{S^{(1)}}]$ with $\mathbf{m}_{S^{(1)}}$ mean of $S^{(1)}$

4. Metropolis-Hastings accept-or-reject. Then fix $S^{(0)}$ and update $S^{(1)}$; iterate them

Other ways of Metropolis adjustments: [Sprungk, Weissmann, Zech 2023]

Dimensional scaling: autocorrelation time at stationarity (same as one-chain counterpart)

Experiments for $\exp\left(-\int_0^1 \frac{1}{2}(\partial_x u(x))^2 + (1-u^2(x))^2 \mathrm{d}x\right)$ discretized into d grid points



For non-negligible acceptance:

- $a - 1 \sim d^{-1/2}$ in stretch move
- $h \sim d^{-1/3}$ in affine-invariant Langevin move

Autocorrelation time

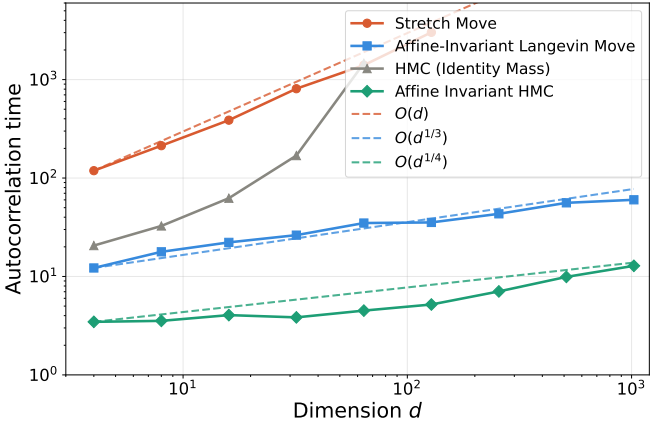
$$O(d) \text{ v.s. } O(d^{1/3})$$

Proved for any Gaussian distributions (and generalizations) [Chen 2025, 2026+]

Note: for this target, function space sampler, such as pCN, could achieve d-independent scaling

Further improvement: Unbiased samplers with affine invariant Hamiltonian flows

Experiments for $\exp\left(-\int_0^1 \frac{1}{2}(\partial_x u(x))^2 + (1-u^2(x))^2 \mathrm{d}x\right)$ discretized into d grid points



Conjugate preconditioned Hamiltonian Monte Carlo (HMC) which is affine invariant

$$\frac{\mathrm{d}\mathbf{x}_i}{\mathrm{d}t} = B_{S^{(1)}}\mathbf{p}_i$$
$$\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = -B_{S^{(1)}}^T \nabla V(\mathbf{x}_i)$$

Leapfrog $h \sim d^{-1/4}$
acceptance non-negligible
autocorrelation time $O(d^{1/4})$
[Chen 2025]

Conjugate-type preconditioning for underdamped Langevin: [Leimkuhler, Matthews, Weare 2018]
Classical HMC optimal scaling result [Beskos, Pillai, Roberts, Sanz-Serna, Stuart 2010]

Current summary

- Affine invariant gradient and Hamiltonian flows to generate proposals in Metropolis-Hastings: improved high-dimensional performance. Goal is to advance `emcee` to benefit astro-community and beyond

  Algorithmically, need adaptations, such as in NUTS, WALNUTS, to tun integration length and stepsize [Hoffman, Gelman 2011], [Bou-Rabee, Carpenter, Kleppe, Liu 2025]

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- These exact MCMC often costly, e.g., for large scale Bayes inverse problems

  Biased approach: no Metropolis-Hasting, just approximate the dynamics

  One theoretical justification of biased over unbiased in high dimensions, if quantities of interest are low-dimensional marginals [Chen, Cheng, Niles-Weed, Weare 2024]

  See also related analysis [Durmus, Eberle 2023] [Bou-Rabee, Schuh 2023] [Cui, Liu, Tong 2024] [Lacker Zhou 2025]

  Many empirical evidence, e.g., [Chada, Leimkuhler, Paulin, Whalley 2023] [Robnik, Cohn-Gordon, Seljak 2024]

Parametric approximation by Gaussians

We can *project* gradient flows in probability space into Gaussian space

- Consider any dynamics in the density space

$$\frac{\partial \rho_t(\theta)}{\partial t} = \sigma_t(\theta, \rho_t)$$

- Write down the dynamics of the mean and covariance

$$\frac{\mathrm{d}m_t}{\mathrm{d}t} = \int \sigma_t(\theta, \rho_t)\theta \mathrm{d}\theta$$
$$\frac{\mathrm{d}C_t}{\mathrm{d}t} = \int \sigma_t(\theta, \rho_t)(\theta - m_t)(\theta - m_t)^T \mathrm{d}\theta$$

- Closure: replace ρ_t in the above RHS by $\rho_{a_t} = \mathcal{N}(m_t, C_t)$, where $a_t = (m_t, C_t)$

Equivalent to Gaussian variational inference $\min_a \mathrm{KL}[\rho_a \| \pi]$ with $a = (m, C)$ and using the projected metric to derive gradient flows in the parameter space

References: Moment closure in variational Kalman filtering [Särkkä, 2007], and in Wasserstein gradient flow [Lambert, Chewi, Bach, Bonnabel, Rigollet 2022]

Projected gradient flows in the Gaussian space: Examples

Direct parameter-space gradient flow versus projection of Wasserstein gradient flows

$$\frac{\mathrm{d}m_t}{\mathrm{d}t} = \mathbb{E}_{\rho_{a_t}}\left[\nabla \log \pi\right] \qquad \frac{\mathrm{d}m_t}{\mathrm{d}t} = \mathbb{E}_{\rho_{a_t}}[\nabla \log \pi]$$

$$\frac{\mathrm{d}C_t}{\mathrm{d}t} = \frac{1}{2}C_t^{-1} + \frac{1}{2}\mathbb{E}_{\rho_{a_t}}[\nabla^2 \log \pi] \qquad \frac{\mathrm{d}C_t}{\mathrm{d}t} = 2I + \mathbb{E}_{\rho_{a_t}}[\nabla^2 \log \pi]C_t + C_t\mathbb{E}_{\rho_{a_t}}\left[\nabla^2 \log \pi\right]$$

Affine invariance: Projection of Fisher-Rao gradient flows

$$\frac{\mathrm{d}m_t}{\mathrm{d}t} = C_t\mathbb{E}_{\rho_{a_t}}[\nabla_\theta \log \pi],$$

$$\frac{\mathrm{d}C_t}{\mathrm{d}t} = C_t + C_t\mathbb{E}_{\rho_{a_t}}[\nabla^2 \log \pi]C_t$$

It is also the *natural gradient* [Amari 1998] flow which is affine invariant

Wasserstein projection [Lambert, Chewi, Bach, Bonnabel, Rigollet 2022] (not affine invariant)

Others related: [Diao, Balasubramanian, Chewi, Salim 2023], [Domke, Garrigos, Gower 2023], [Liero, Mielke, Tse, Zhu 2025], [Petit-Talamon, Lambert, Korba 2025], ...

Affine invariance leads to robustness to anisotropy

A numerical experiment on anisotropic distributions

- 2D potential with $\theta = (\theta^{(1)}, \theta^{(2)})$
- $\pi \propto \exp(-V)$
- $V = \frac{(\sqrt{\lambda}\theta^{(1)} - \theta^{(2)})^2}{20} + \frac{(\theta^{(2)})^4}{20}$
- $\lambda = 0.01,\ 0.1,\ 1$

Note: V is convex

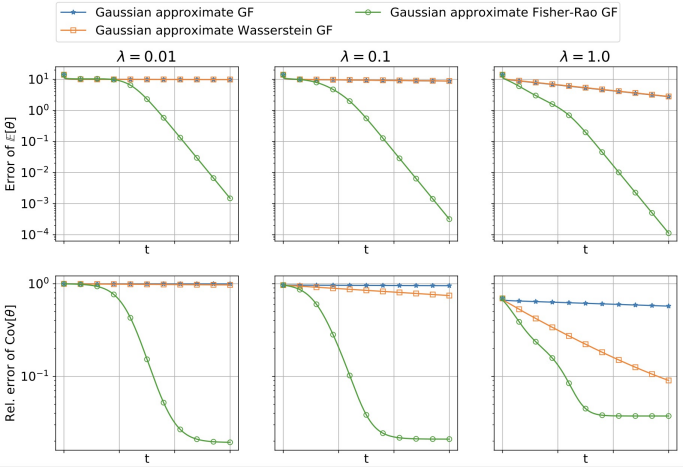


Figure: t axis is from $t = 0$ to 15. Convergence rate of Gaussian approximate Fisher-Rao gradient flows not influenced by values of λ

Theory regarding convergence rate dependence on condition number

Gaussian targets: global convergence rate

If $\pi = \mathcal{N}(m_\star, C_\star)$, and $C_0 = \lambda_0 I, \lambda_0 > 0$, then

$$\|m_t - m_\star\|_2 = \mathcal{O}(e^{-t}), \quad \|C_t - C_\star\|_2 = \mathcal{O}(e^{-t})$$

Rate is independent of condition number of $C_\star$

Log-concave targets: local convergence rate

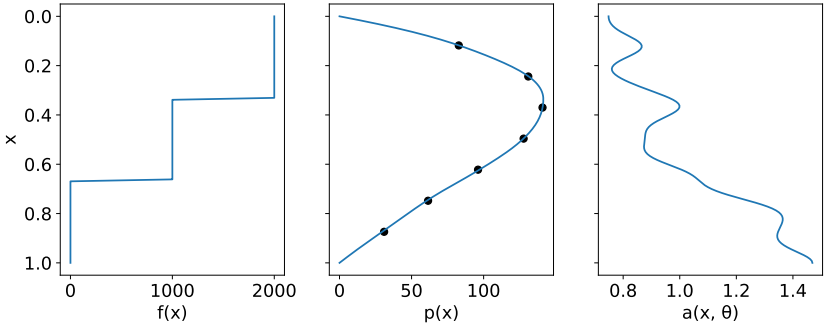
Assume $\alpha I \preceq -\nabla^2 \log \pi \preceq \beta I$. For $d = 1$, let $\lambda_{\star,\max} < 0$ denote the largest eigenvalue of the linearized Jacobian matrix of the flow around the optimal $a_\star$. We have

$$\lambda_{\star,\max} \leq -\frac{1}{(7 + \frac{4}{\sqrt{\pi}})(1 + \log \kappa)}$$

where $\kappa = \beta/\alpha$ is the condition number

1D Darcy flow Bayesian inverse problem example

Recover $a(x, \theta)$ from noisy values of p, equation $-\nabla \cdot (a \nabla p) = f$, and known f



$\log a(x, \theta)$ is modeled as Gaussian process (GP), parametrized through a truncated Karhunen-Loève (KL) expansion (first 16 parameters). Details in paper. Goal is to sample the posterior of these parameters under the noisy observations $p(x_i) + \xi_i, 1 \leq i \leq 7$

1D Darcy flow experiments: Error curves for different methods

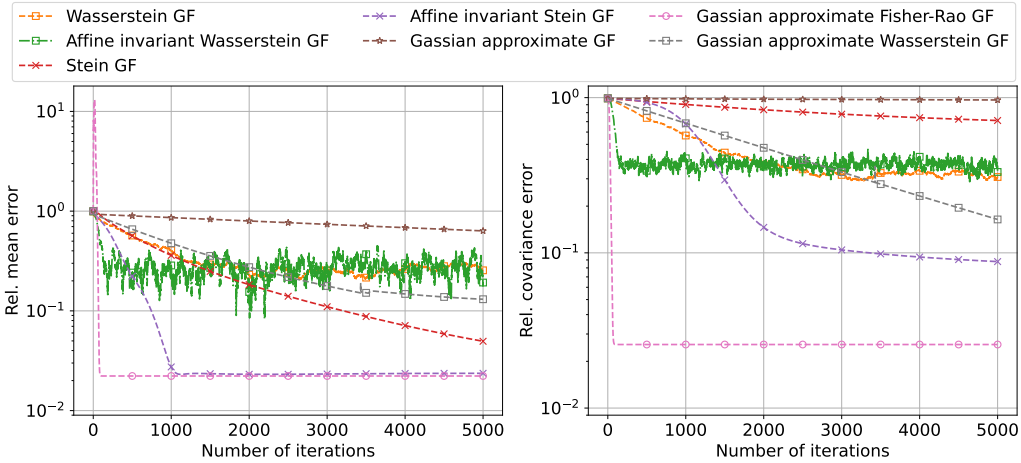


Figure: Relative error of mean and covariance of gradient flows

Gaussian mixtures and interacting density gradient flows

Interacting affine-invariant (Fisher-Rao) gradient flows over Gaussian mixtures

$$\dot{m}_{t,k} = -C_{t,k} \int \mathcal{N}(\theta; m_{t,k}, C_{t,k}) \Big(\nabla_\theta \log \rho_t - \nabla_\theta \log \pi \Big) \mathrm{d}\theta$$

$$\dot{C}_{t,k} = -C_{t,k} \Big(\int \mathcal{N}(\theta; m_{t,k}, C_{t,k}) \big(\nabla_\theta^2 \log \rho_t - \nabla_\theta^2 \log \pi \big) \mathrm{d}\theta \Big) C_{t,k}$$

$$\dot{w}_{t,k} = -w_{t,k} \int (\mathcal{N}(\theta; m_{t,k}, C_{t,k}) - \rho_t)(\log \rho_t - \log \pi) \mathrm{d}\theta$$

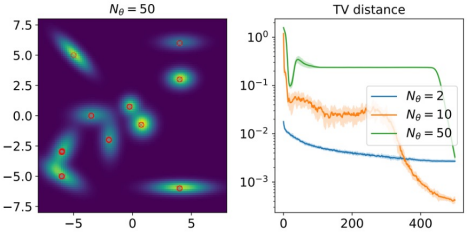
Here $\rho_t(\theta) = \sum_{k=1}^K w_{t,k} \mathcal{N}(\theta; m_{t,k}, C_{t,k})$. Numerically need to be careful

This can be seen as the natural gradient flow for optimizing

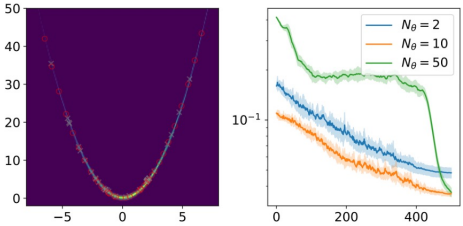
$$\min_{w_k, m_k, C_k} \mathrm{KL}[\sum_{k=1}^K w_k \mathcal{N}(\theta; m_k, C_k) \| \pi]$$

with diagonal approximations of the Fisher information matrix [Lin, Khan, Schmidt 2020]

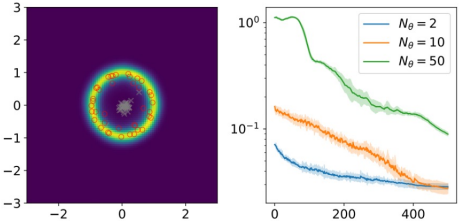
Experiments: affine invariant gradient flows over Gaussian mixtures (iterations $O(10^2)$)



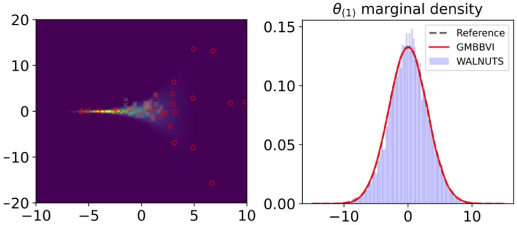
Multimodal Gaussian mixtures



Curved Rosenbrock distributions



Concentrated ring distributions

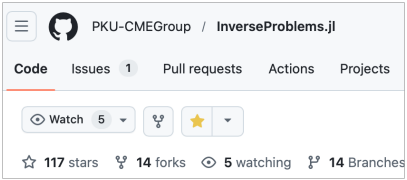


Multiscale funnel distributions

2D Darcy flow PDE inverse problems: Gaussian mixtures work efficiently

This concerns a symmetry setting, leading to two posterior modes

$\log a(\cdot, \theta)$ is modeled by GPs, using truncated KL expansions with first $d = 128$ coefficients



check the repo!

[Chen, Huang, Huang, Reich, Stuart 2024], [Che, Chen, Huang, Mao, Wang 2025], [Che, Chen, Huan, Huang, Wang 2026]

Figure: Truth a, and recovery by 3-mode Gaussian mixtures affine invariant gradient flow approach (derivative-free implementation)

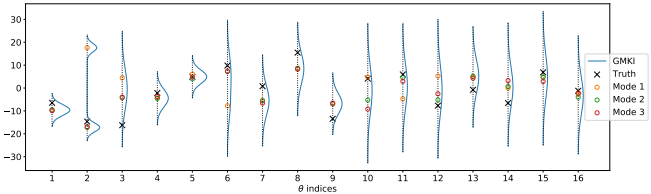


Figure: For the first 16 KL coefs: Truth expansion parameters (black crosses), and mean estimations of for each mode (circles) and the associated marginal distributions at the 30-th iteration

Conclusions: affine invariant samplers and gradient/Hamiltonian flows

- Affine invariant samplers are simple and routinely used in applications (Goodman-Weare and `emcee` package) but can suffer from high dimensions
  - $O(d)$ autocorrelation time at stationarity for Gaussian targets
  - scaling limits as affine invariant gradient flows
- Improved dimensional scaling using affine invariant gradient flows, and Hamiltonian flows as proposals in unbiased Metropolis-Hastings
  - $O(d^{1/3})$ and $O(d^{1/4})$ autocorrelation time at stationarity for Gaussian targets
  - towards an improved toolkit for exact Bayesian calibration in astrophysics and beyond
- Further scalability by approximate, biased methods with variational inference
  - affine invariant gradient flows over Gaussian space: global convergence rate independent from κ for Gaussian targets, and local convergence rate depends on $O(\log \kappa)$ for log-concave targets, where κ is condition number
  - over Gaussian mixtures: towards a simple, generally applicable, and derivative-free toolkit for posterior approximation in large-scale Bayesian inverse problems

References

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Thank you!

Back-Up Slides

Gaussian mixture experiments

Consider the 2d Darcy flow problem ($\theta \in \mathbb{R}^{128}$)

$$-\nabla \cdot (a(x,\theta)\nabla p(x)) = f(x) = 1000 \sin\big(4\pi x_{(2)}\big) \qquad x \in D$$
$$p(x) = 0 \qquad x \in \partial D$$

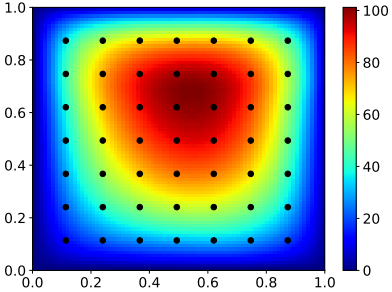

Figure: The reference pressure field p and observations $\frac{p([x_{(1)},x_{(2)}]^T)+p([1-x_{(1)},x_{(2)}]^T)}{2}$ at 28 equidistant points (solid black dots) and their mirroring points (empty black dots).

Ensemble affine invariant sampler [Goodman, Weare 2010]

Affine invariance in MCMC is widely achieved by ensemble samplers

Ensemble samplers: at a discrete m-step

$$(\mathbf{x}_1(m+1), \mathbf{x}_2(m+1), \ldots, \mathbf{x}_N(m+1)) = R(\mathbf{x}_1(m), \mathbf{x}_2(m), \ldots, \mathbf{x}_N(m), \pi)$$

Target distribution is π^N in the product space

Affine invariance: for any invertible affine transformation $\mathbf{y} = \phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$

$$(\mathbf{x}_1, \ldots, \mathbf{x}_N) \xrightarrow{\phi} (\mathbf{y}_1, \ldots, \mathbf{y}_N) = (A\mathbf{x}_1 + \mathbf{b}, \ldots, A\mathbf{x}_N + \mathbf{b}),$$

it holds that

$$(\mathbf{y}_1(m+1), \mathbf{y}_2(m+1), \ldots, \mathbf{y}_N(m+1)) = R(\mathbf{y}_1(m), \mathbf{y}_2(m), \ldots, \mathbf{y}_N(m), \phi\#\pi)$$

Ensemble affine invariant sampler [Goodman, Weare 2010]

Affine invariance in MCMC is widely achieved by ensemble samplers

Stretch move sampler

- Ensemble at step $m \in \mathbb{N}$ $(\mathbf{x}_1(m), \mathbf{x}_2(m), ..., \mathbf{x}_N(m))$
- Stretch move randomly selects two $\mathbf{x}_i(m)$ and $\mathbf{x}_j(m)$

$$\tilde{\mathbf{x}}_i(m+1) = \mathbf{x}_j(m) + Z(\mathbf{x}_i(m) - \mathbf{x}_j(m))$$

  where density of Z satisfies $g(z) \propto \frac{1}{\sqrt{z}}$ for $z \in [a^{-1}, a]$; recommended $a = 2$
- Accept this proposal with probability

$$\min\{1, Z^{d-1}\frac{\pi(\tilde{\mathbf{x}}_i(m+1))}{\pi(\mathbf{x}_i(m))}\}$$

- Affine invariant since based on relative locations
- Used routinely in Bayesian applications: `emcee` package (>10k citations)

  [Foreman-Mackey, Hogg, Lang, Goodman 2013]